Final Report

A. Report Information

- **Course:** Social Network Analysis
- **Project:** A10. Author Collaborator Finder (ArXiv Co-auth)
- **Duration:** 09/2025 - 11/2025
- **Group**: 2
- **Members:**

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B. Report Content

Abstract

This project studies the problem of link prediction in scientific collaboration networks using the ArXiv Hep-Th dataset. We construct and analyze the co-authorship graph to examine structural properties such as degree distribution and clustering coefficients. To suggest potential future collaborators, we implement and evaluate classical proximity-based methods, including Common Neighbors, Adamic–Adar, Resource Allocation, and Preferential Attachment. Furthermore, we investigate the role of community structure by applying the Louvain algorithm and proposing a hybrid approach that combines local similarity with community overlap. Experimental results under a time-aware train–test split reveal distinct performance regimes: while neighbor-based heuristics significantly outperform Preferential Attachment, our proposed additive hybrid framework yields the highest accuracy, successfully synthesizing local topology with global community signals to surpass individual baselines. Finally, a web-based prototype is developed to demonstrate top-K co-author recommendations.

1. Introduction

1.1 Motivation

Collaboration is a cornerstone of modern scientific discovery, particularly in specialized fields like theoretical physics where complex problems demand diverse expertise. While identifying suitable collaborators is critical for impactful research, scholars often rely on limited immediate social circles, missing potential partnerships. The exponential growth of digital repositories such as ArXiv offers a data-driven opportunity to overcome this limitation by modeling scientific interactions as co-authorship networks. Analyzing these networks allows us to investigate the structural mechanisms of collaboration evolution - such as community dynamics and triadic closure. Consequently, the problem of link prediction becomes not only a theoretical inquiry into network topology but also a practical foundation for building recommender systems that facilitate meaningful scientific connections.

1.2 Problem Statement

This study addresses the challenge of predicting future co-authorships within the ArXiv Hep-Th collaboration network. The primary objective is to forecast likely research partnerships based on historical interaction patterns. Despite the breadth of existing literature, significant gaps remain that limit practical applicability. First, many conventional approaches utilize static network snapshots, ignoring the critical temporal dynamics inherent in scientific evolution. Second, standard evaluations often suffer from temporal bias by randomly splitting edges rather than respecting chronological order. Third, few studies translate predictive models into interactive systems with transparent explainability - a crucial requirement for user trust in real-world scenarios. Consequently, this project aims to bridge these gaps by constructing a rigorous time-aware dataset, benchmarking proximity-based algorithms under a temporal split, and deploying an interpretable collaborator recommendation prototype.

1.3 Objectives & Scope

Objectives: The primary goal of this project is to develop an explainable collaborator recommendation system based on temporal link prediction. To achieve this, we first aim to construct a rigorous temporal co-authorship network from the ArXiv Hep-Th metadata (1992–2003), enabling the detailed analysis of key topological features such as connectivity and clustering coefficients. Building on this structural foundation, we seek to implement and evaluate a suite of predictive models, benchmarking classical proximity heuristics (e.g., Common Neighbors, Adamic–Adar) against community-aware hybrid approaches. Crucially, we intend to establish a strict time-aware validation protocol - training on historical data and testing on future interactions - to prevent information leakage and ensure realistic forecasting. Finally, the project aims to deploy the top-performing models within an interactive Gradio prototype that delivers interpretable top-K recommendations enriched with justification paths, thereby bridging the gap between graph theory and practical utility.

Scope:

The scope of this project is strictly defined to ensure a focused examination of structural link prediction. We concentrate specifically on undirected, weighted co-authorship graphs derived from the Hep-Th metadata, utilizing unsupervised heuristic algorithms and Louvain community detection as the core analytical engines. The evaluation is grounded in AUC and Average Precision metrics, with practical interpretation provided via local topology visualization in a Gradio-based prototype.

Conversely, certain advanced methodologies remain out of scope for this foundational study; these include supervised machine learning models (e.g., Random Forests, GNNs), complex node embedding techniques (e.g., Node2Vec), and semantic content analysis of paper abstracts. Furthermore, author disambiguation is limited to rule-based normalization, excluding large-scale entity resolution or cross-domain analysis beyond the theoretical physics category.

1.4 Contributions

This project contributes a systematic evaluation of graph-based link prediction techniques applied to scientific collaboration, bridging the gap between theoretical network analysis and practical recommendation systems. First, we conducted a comprehensive benchmarking of classical heuristics on the ArXiv Hep-Th network, revealing the superior performance of local neighbor-based methods over popularity-driven approaches in this domain. Second, we investigated the limits of structural prediction by determining global research communities via the Louvain algorithm and developing an optimized additive hybrid model that effectively synthesizes local similarity with community overlap, demonstrating a proven performance gain over individual baselines. Third, addressing the need for transparency, we engineered an interactive prototype that transcends black-box predictions by providing verifiable justifications - specifically, shared neighbors and connectivity paths - to enhance user trust.

These efforts culminated in a complete research package comprising a cleaned temporal co-authorship dataset, a reproducible codebase for heuristic evaluation, a rigorous quantitative analysis report, and a functional Gradio-based web application for real-time, explainable co-author discovery.

2. Related Work

2.1 Background Knowledge

To facilitate the subsequent analysis, we formally define the problem and introduce the necessary mathematical notations and algorithms utilized in this study.

- Network Definitions: We model the co-authorship network as an undirected, weighted graph $G = (V, E)$, where V represents the set of authors and E represents the set of collaborative relationships. An edge $(u, v) \in E$ exists if authors u and v have co-authored at least one paper. The weight $w(u, v)$ denotes the frequency of collaboration.
    - Neighbors: Let $Gamma(u)$ be the set of neighbors of node u, defined as

      $$Gamma(u) = \{ v \in V \mid (u, v) \in E \}.$$

    - Degree: The degree of node u, denoted k_u, is the number of neighbors,

      $$k\_u = |Gamma(u)|.$$

- Link Prediction Metrics: We focus on unsupervised proximity-based methods that assign a similarity score $score(u, v)$ to pairs of non-connected nodes. The specific metrics implemented in this study are:
    - Common Neighbors (CN): Assumes that two authors with many mutual collaborators are likely to collaborate.

      $$S\_\{CN\}(u, v) = |Gamma(u) \cap Gamma(v)|$$

    - Adamic–Adar (AA): Refines CN by penalizing common neighbors with high degrees, giving more weight to shared neighbors who are themselves less connected.

      $$S\_\{AA\}(u, v) = \Sigma\,(1 \,/\, log(k\_z)\ for\ z \in Gamma(u) \cap Gamma(v))$$

    - Resource Allocation (RA): Similar to AA but penalizes high-degree common neighbors more heavily.

      $$S\_\{RA\}(u, v) = \Sigma\,(1 \,/\, k\_z\ for\ z \in Gamma(u) \cap Gamma(v))$$

    - Preferential Attachment (PA): Based on the premise that highly connected authors are more likely to form new ties (“rich get richer”).

$$S_{PA}(u, v) = k_u \times k_v$$

- Community Detection: To capture global structural information, we employ the Louvain Algorithm, a modularity maximization method. It partitions the graph into communities where the density of edges within communities is significantly higher than expected by chance. This allows us to determine if two authors belong to the same research cluster, providing a global signal to complement local proximity metrics.

2.2 Prior Work Comparison & Limitations

Link prediction in co-authorship networks has evolved significantly, shifting from fundamental topological heuristics to advanced machine learning paradigms. Early foundational studies demonstrated that simple neighbor-based metrics, such as Common Neighbors and Adamic–Adar, serve as powerful baselines by exploiting local structural homophily *[6]*. Recent advancements have increasingly integrated auxiliary data to enhance predictive performance; for instance, Hassan & Hasan (2025) introduced supervised frameworks leveraging rich author node-based features *[1]*, while Chuan et al. (2018) and Yuliansyah et al. (2025) combined structural properties with semantic content like article keywords to formulate hybrid similarity metrics *[3]*, *[5]*. Furthermore, complex architectures such as the multilayer analysis by Wang et al. (2023) and Graph Neural Networks on the OGB collaboration dataset have achieved state-of-the-art results by modeling high-order network dependencies *[2]*, *[7]*.

Despite these methodological strides, existing literature presents notable limitations that hinder practical deployment. A primary concern is the trade-off between complexity and interpretability; while advanced deep learning and hybrid semantic models offer improved accuracy, they often operate as “black boxes”, failing to provide transparent justifications for their recommendations - a critical requirement for user trust. Additionally, many studies suffer from temporal naïvety, evaluating models on static snapshots or randomized train-test splits rather than strictly strictly adhering to chronological order, which can lead to information leakage and overoptimistic performance estimates *[2]*. Finally, there remains a gap in translating theoretical metrics into accessible tools, as few works develop interactive prototypes for real-time, explainable collaboration discovery *[4]*, *[6]*. Addressing these gaps, our study revisits classical heuristics within a rigorous time-aware framework, augmented by community detection, to deliver a transparent and practically applicable recommender system.

2.3 Positioning of This Work

This study situates itself within the domain of graph-based link prediction for scientific collaboration, distinguishing itself by prioritizing interpretability and transparency over purely performance-driven, black-box architectures. Unlike recent trends that leverage opaque deep learning embeddings, our approach revisits and rigorously benchmarks classical structural methods to establish a transparent baseline for collaboration recommendation.

Our work advances beyond isolated algorithm evaluations by systematically comparing diverse proximity heuristics (e.g., Common Neighbors, Adamic-Adar) alongside community-aware models on a unified, real-world Hep-Th dataset. By synthesizing local structural signals with global community membership, we demonstrate how multi-scale graph information elucidates the mechanisms of collaboration formation.

Crucially, we address the “interpretability gap” prevalent in existing literature by integrating plain-text justifications - specifically, shared neighbors and shortest paths - directly into the recommendation workflow. This shifts the focus from abstract probability scores to actionable insights. In summary, this project offers a balanced perspective between predictive accuracy and model transparency, proving that lightweight, fundamental graph techniques can yield competitive results and practical utility without the computational overhead of complex neural frameworks.

3. Methodology

3.1 Overview of Approach

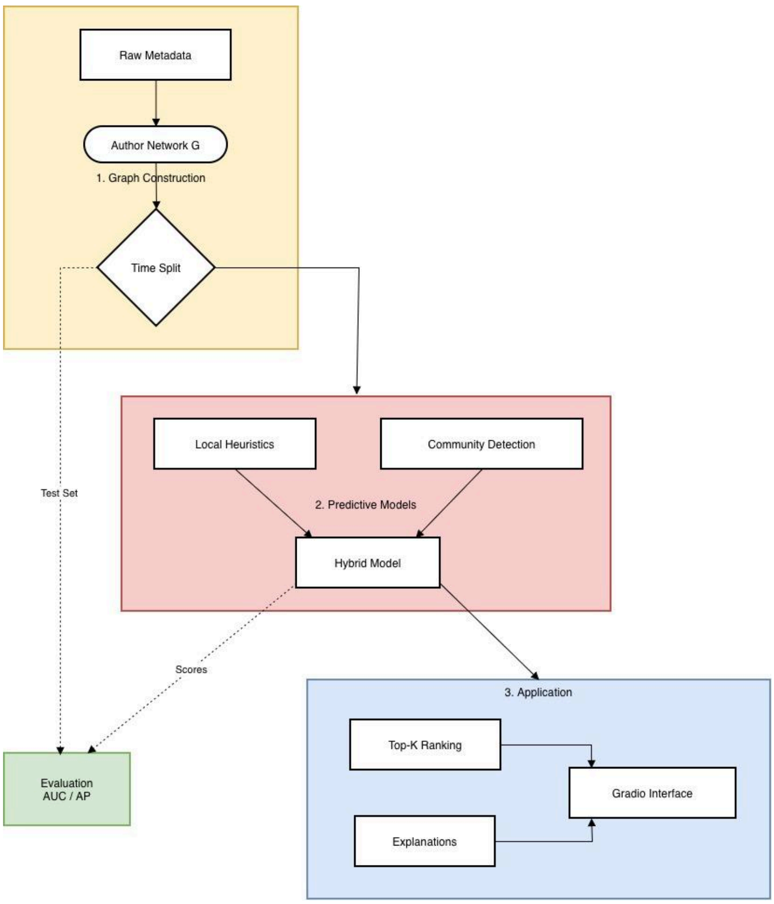


Figure 1: High-level architectural pipeline of the proposed co-author recommendation system.

As illustrated in the system architecture diagram **(Figure 1)**, the pipeline is organized into three main functional modules designed to transform raw bibliographic data into an explainable recommendation system. The workflow begins with Graph Construction, where raw metadata is ingested to build the initial Author Network (G). Crucially, a Time Split mechanism partitions this data, creating a training graph for modelling while rigorously preserving future interactions as a separate Test Set (indicated by the dotted line) for unbiased benchmarking. Following this, the Predictive Models module extracts features using two parallel approaches: Local proximity heuristics (capturing neighborhood similarity) and Community detection (capturing global structure). These components feed into a Hybrid model that synthesizes local and global signals, producing predictive Scores that are validated against the Test Set via AUC and Average Precision metrics in the Evaluation block. Finally, the pipeline culminates in the Application Interface module, which encapsulates the trained models within a user-facing Gradio Interface. This system not only generates Top-K recommendations but distinctly provides Explanations - such as shared neighbors or justification paths - offering transparency alongside prediction.

3.2 Data

3.2.1 Source and Acquisition

This study utilizes the ArXiv Hep-Th (High Energy Physics – Theory) dataset, comprising metadata of research papers from 1992 to 2003. The raw data consists of semi-structured .abs files, from which we programmatically extract key attributes including paper identifiers, publication dates, titles, and author lists. This corpus serves as the foundation for constructing the temporal co-authorship network.

3.2.2 Preprocessing and Cleaning

To transform noisy raw text into a rigorous graph structure, we implement a multi-stage preprocessing pipeline:

- Name Normalization: We employ regex-based spatial filtering to standardize author names. This includes handling LaTeX accents (e.g., converting "u to ue) and unifying spacing variations. For instance, “H. D. Dahmen” and “H.D.Dahmen” are merged into a single unique identifier to match their publication history correctly.
- Ambiguity Handling: We adopt a conservative disambiguation strategy where unique normalized name strings are treated as distinct authors. While this prevents the incorrect merging of distinct researchers with similar names (e.g., “J. Smith” vs. “J. A. Smith”), it accepts a known limitation: distinct authors sharing the exact same name (homonyms) will be merged into a single node. Given the metadata-only nature of the dataset, we prioritize this simple, deterministic rule over complex disambiguation heuristics that often introduce unpredictable errors.
- Entity Filtering: Malformed entries, such as page counts masquerading as names or single-token scraps, are systematically removed.

3.2.3 Network Construction

The co-authorship graph $G = (V, E)$ is constructed as an undirected, weighted network:

- Nodes V: Represent individual authors. Crucially, we retain isolated authors (those with single-author papers only) to preserve the true population distribution.
- Edges E: An edge connects two authors if they have co-authored at least one paper.
- Edge Attributes: Each edge carries a weight representing the total number of collaborations and a first_year timestamp marking their initial partnership.
- Characteristics: The resulting graph is sparse and exhibits a heavy-tailed degree distribution, typical of social collaboration networks.

3.2.4 Temporal Evaluation Protocol

To simulate a realistic forecasting scenario and avoid “future leakage”, we partition the graph temporally rather than randomly:

- Training Set: Comprises all edges formed on or before a split year T (e.g., 1998).
- Test Set: Contains new collaborations formed after T.
- Cold-Start Constraint: Evaluation is strictly restricted to pairs of authors who are already present in the training graph. This focus ensures the model predicts collaborations between established researchers rather than new entrants (cold-start problem).
- Negative Sampling: We generate non-collaborating pairs uniformly from the active training node pool to maintain a balanced evaluation framework for metric calculation (AUC/AP).
- Prediction Scope (New Links Only): The system is designed to discover novel collaborations. We strictly exclude any pair of authors who have already co-authored a paper in the training period. The model only predicts links for pairs that currently have no direct connection in the graph $G\_\{Train\}$.

3.2.5 Processed Data Schema

Table 1: Schema of the Processed Co-authorship Dataset.

Column Name	Type	Description
author1	String	Name of the first author (normalized).
author2	String	Name of the second author (normalized).
first_year	Integer	The earliest year these

		two authors collaborated (used for temporal splitting).
weight	Integer	The total number of papers co-authored within the dataset period (1992-2003).

The final dataset used for model training and evaluation is structured as a relational table of co-authorship edges with the following schema **(Table 1)**

3.3 System Design

Our system is architected around three core predictive modules designed to capture complementary aspects of scientific collaboration: local topology, global community structure, and a hybrid synthesis.

3.3.1 Local Proximity Module

This module computes similarity scores based on immediate neighborhood overlap. We select four representative heuristics that offer a trade-off between simplicity and structural awareness:

- Common Neighbors (CN): Serving as the baseline, this metric quantifies the raw volume of shared collaborators.
- Adamic-Adar (AA) & Resource Allocation (RA): These metrics refine CN by penalizing high-degree common neighbors, effectively down-weighting "hub" authors who collaborate indiscriminately. This is crucial in scientific networks where connecting via a famous professor is less significant than connecting via a specific domain expert.
- Preferential Attachment (PA): This metric models the "rich-get-richer" phenomenon, assuming collaboration is driven solely by individual popularity rather than shared context. It serves as a contrasting baseline to neighbor-based methods.

3.3.2 Community-Aware Module

To capture global structural patterns beyond local neighborhoods, we employ the Louvain Algorithm. This iterative method optimizes modularity to partition the network into dense research communities $C\_1,\ C\_2,\ \ldots,\ C\_k$.

- Justification: Collaboration is often homophilous with respect to research topics. By identifying community boundaries, we can boost predictions for author pairs residing within the same semantic cluster.

- Scoring: We define a binary community score $S\_\{Comm\}(u, v) = 1$ if u and v belong to the same partition, and 0 otherwise.

3.3.3 Hybrid Inference Engine

We propose a hybrid scoring mechanism to integrate local and global signals. The rationale is that while neighborhoods drive immediate collaboration, community membership provides a broader "topic constraint". The hybrid score is defined as:

$$S\_\{Hybrid\}(u, v) = S\_\{Local\}(u, v) + (\alpha . S\_\{Comm\}(u, v))$$

- Where $S\_\{Local\}$ is the base heuristic (e.g., Adamic-Adar).
- $S\_\{Comm\}$ is the binary community indicator (1 if same community, 0 otherwise).
- α is a tunable additive bias parameter (set to α =2.0 in our experiments) to explicitly reward intra-community potential links.

We adopt an additive formulation over multiplicative alternatives based on empirical comparison. The additive approach offers two critical advantages. First, zero-score rescue: when $S\_\{Local\}$ equals 0 due to absent common neighbors, multiplicative models remain at 0 regardless of community membership, whereas additive models assign score alpha, enabling discovery of topologically distant but thematically aligned pairs. Second, independent signal contribution: additive combination allows local and global signals to contribute proportionally without one dominating the other. The parameter alpha equals 2.0 was selected via grid search over the set 0.5, 1.0, 2.0, 5.0, 10.0, balancing community influence without overwhelming fine-grained local distinctions.

The hybrid scoring operates across four distinct regimes. When both signals align (for example, AA equals 2.8 and the same community), the score is 4.8, indicating high-confidence prediction. When only local signal exists (AA equals 1.5 but different communities), the score is 1.5, representing moderate confidence based solely on neighborhood overlap. Critically, when AA equals 0 but authors share a community, the score is 2.0, rescuing links that pure local methods would miss entirely. When both signals are absent, the score remains 0, correctly indicating no evidence for collaboration.

This design addresses the fundamental challenge of network sparsity. In our dataset, 97.7 percent of test edges have zero common neighbors, rendering local methods ineffective for the majority of predictions. The hybrid model's community signal rescues these zero-score cases, improving recall while maintaining interpretability through transparent score decomposition.

3.3.4 Recommendation & Explanation Layer

The final output is generated by ranking candidate pairs according to $S\_\{Hybrid\}$. To ensure interpretability, the system retrieves qualitative evidence for top-ranked predictions:

- Explanation 1: List of specific Shared Neighbors (the "bridge" authors).

- Explanation 2: Existence of a Shortest Path $(length \leq 4)$ in the training graph, illustrating the closeness of the two researchers.

3.4 Training & Implementation Details

3.4.1 Graph Representation & Sampling

We model the collaboration network as an undirected, weighted graph, where edge weights correspond to the frequency of co-authored papers. To manage computational cost during evaluation, we do not compute scores for all possible non-existing links $(O(N^2))$. Instead, we adopt a candidate sampling strategy:

- For Top-K Recommendation: We restrict the search space to the 2-hop neighborhood of the query author (neighbors of neighbors), which covers the most likely future collaborators.

- For Metric Evaluation: We generate a fixed set of negative samples (non-edges) equal in size to the positive test set, ensuring a balanced 1:1 ratio for fair AUC calculation.

3.4.2 Evaluation Protocol

We employ a strict Time-Aware Validation scheme to prevent look-ahead bias.

- Training Phase: The graph is constructed using only collaborations initiated on or before the split year T.

- Testing Phase: “Ground truth” links are defined as new collaborations forming after year T between authors already present in the training set.

- No iterative “training” (in the machine learning sense) is performed, as the heuristics are parameter-free. However, the evaluation serves to benchmark the discriminative power of each topological feature.

3.4.3 Algorithms & Hyperparameters

We evaluate a specific set of parameter-free proximity metrics: Common Neighbors (CN), Adamic–Adar (AA), Resource Allocation (RA), and Preferential Attachment (PA). For the Hybrid model, we integrate the modularity partitions from the Louvain algorithm (default resolution = 1.0, randomized seed for reproducibility) to boost local scores when authors share community membership.

3.5 Evaluation Protocol

3.5.1 Validation Framework

To rigorously assess predictive performance, we employ the time-aware validation framework described in Section 3.2.4. The dataset is partitioned chronologically at the split threshold T, ensuring that the model is evaluated strictly on future collaborations ("forecasting") rather than missing links. We maintain a balanced 1:1 ratio between positive test edges and randomly sampled negative non-edges to ensure fair AUC calculation.

3.5.2 Quantitative Metrics

We utilize two standard metrics for binary link classification:

- Area Under the ROC Curve (AUC): Evaluates the model's global discriminative power - specifically, the probability that a random future collaboration is scored higher than a random non-collaboration. It is robust to class imbalance.

- Average Precision (AP): Summarizes the precision-recall trade-off, placing greater emphasis on the top-ranked predictions. This metric is particularly relevant for recommendation tasks where the quality of the highest-scored candidates is paramount.

3.5.3 Comparative Benchmarks

We systematically benchmark three categories of predictive signals:

- Local Topology: Proximity heuristics including Common Neighbors (CN), Adamic–Adar (AA), Resource Allocation (RA), and Preferential Attachment (PA).

- Global Structure: A community-based baseline derived from Louvain partitions, assigning binary scores based on shared community membership.

- Hybrid Approaches: Composite models that integrate local proximity with global community constraints.

3.5.4 Qualitative Assessment

Beyond numerical metrics, we conduct a qualitative assessment via a Top-K Recommendation prototype. This interactive setting allows for the inspection of specific candidate suggestions and their associated justifications (shared neighbors, paths), verifying whether high-scoring predictions align with intuitive collaborative logic.

Prediction Logic in Prototype: It is important to note that the prototype operates on the Training Graph $(G\_\{train\})$ constructed from data up to the split year $T = 1998$. When a user queries an author, the system simulates a forecasting scenario: it utilizes only historical connections (pre-1998) to recommend collaborators for the subsequent period (post-1998). This design allows us to verify the recommendations against actual historical records (the "Ground Truth" test set) to confirm accuracy.

3.5.5 Tie-Breaking Protocol

Network sparsity induces severe score degeneracy in our evaluation. Specifically, 97.7 percent of test edges have zero common neighbors, resulting in 99.4 percent of edge pairs receiving identical scores under neighbor-based methods. Community detection produces only two unique values (0 or 1), yielding 100 percent tied scores. These ties create ranking ambiguity when computing AUC and AP, as standard implementations must arbitrarily order identically-scored edges.

To ensure robust and reproducible evaluation, we employ stochastic tie-breaking with multiple runs. For each method, we add a uniform random noise epsilon drawn from U(0, 10 to the power of negative 10) to all scores. This noise magnitude is negligible relative to typical score ranges (10 orders of magnitude smaller), preserving true rankings for non-tied edges while breaking ties fairly. We repeat evaluation 10 times with different random seeds and report mean plus or minus standard deviation. The consistently small standard deviations (less than 0.005 for all methods) confirm result stability despite tie-breaking randomness, validating the reliability of our performance estimates.

4. Experiments and Results

4.1 Experimental Settings

4.1.1 Dataset Description Experiments

Experiments rely on the ArXiv Hep-Th dataset processed as detailed in Section 3.2.3. The final collaboration graph comprises 13,003 authors (nodes) and 23,847 collaborative relationships (edges). The network exhibits typical characteristics of scientific communities: high sparsity (avg. degree ~3.67) and significant fragmentation (over 2,000 connected components), posing a realistic challenge for prediction algorithms.

4.1.2 Preprocessing & Partitioning

Raw text metadata is cleaned via a regex-based normalization pipeline to standardize author identities. We construct the network from the processed edge list and apply a Time-Aware Split:

- Training Set: Edges formed on or before the split year (e.g., $T = 1998$), consisting of 13,162 edges.
- Test Set: 10,685 positive edges formed strictly after year T.
- To ensure robustness, the analysis is performed on the entire graph structure rather than restricting it solely to the giant connected component, testing the algorithms' ability to handle fragmented network topologies.

4.1.3 Methods & Baselines

The comparative study evaluates:

- Local Heuristics: Common Neighbors (CN), Adamic–Adar (AA), Resource Allocation (RA), and Preferential Attachment (PA).

- Global Strategies: Community membership indicators derived from the Louvain Algorithm.
- Hybrid Models: We implemented hybrid variants for all four base heuristics (denoted as HYB_CN, HYB_AA, HYB_RA, HYB_PA). Each variant amplifies the local score if the author pair shares the same community. After a preliminary comparison, HYB_AA (Hybrid Adamic-Adar) was selected as the representative model for the final discussion due to the robustness of the underlying Adamic-Adar metric in sparse networks.

4.1.4 Configuration & Environment

- Hyperparameters: Heuristics are parameter-free. Louvain uses default resolution (1.0) with a fixed seed. The Explainer module is configured to search for shortest paths up to a cutoff depth of 4 and returns Top-K=10 recommendations.
- Evaluation: Balanced AUC/AP calculation with 1:1 negative sampling derived from the active training node pool.
- Implementation Stack: Code is written in Python 3, utilizing NetworkX for graph operations, python-louvain for clustering, and Gradio for the prototype interface. Experiments are executed on a standard Google Colab CPU instance.

4.2 Quantitative Results

Table 2: Link Prediction Performance (mean ± std over 10 runs with tie-breaking).

Method	AUC	AP
Common Neighbors (CN)	0.5221 ± 0.0045	0.5511 ± 0.0036
Adamic-Adar (AA)	0.5221 ± 0.0045	0.5512 ± 0.0037
Resource Allocation (RA)	0.5221 ± 0.0045	0.5512 ± 0.0037
Preferential Attachment (PA)	0.1945 ± 0.0007	0.4083 ± 0.0014
Louvain	0.5270 ± 0.0045	0.5534 ± 0.0035
Hybrid CN (HYB_CN)	0.5316 ± 0.0046	0.5659 ± 0.0036
Hybrid AA (HYB_AA)	0.5316 ± 0.0046	0.5659 ± 0.0036
Hybrid RA (HYB_RA)	0.5659 ± 0.0036	0.5658 ± 0.0036

Hybrid PA (HYB_PA)	0.1965 ± 0.0007	0.4098 ± 0.0014

Failure of Preferential Attachment: Notably, PA performs poorly with an AUC of only ~0.19, significantly worse than random guessing (AUC=0.5). This indicates a strong “anti-preferential” trend in the test period: new collaborations were not primarily formed between high-degree hubs. Instead, researchers likely preferred connecting with peers of similar standing or within specific niche sub-communities rather than chasing “stars”.

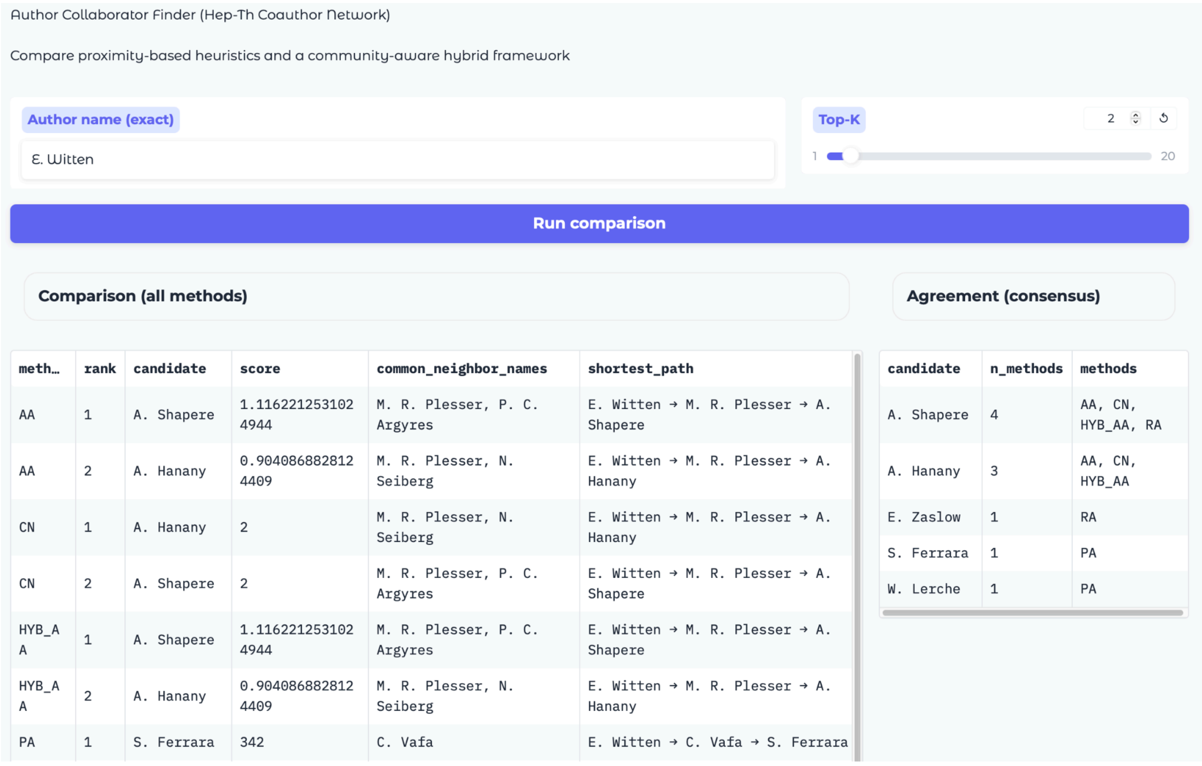
Local Heuristics: Neighbor-based methods (CN, AA, RA) show consistent performance $(AUC \approx 0.52)$. While modest, they significantly outperform the popularity-based baseline, confirming that triadic closure (friends of friends collaborating) remains a valid, albeit weak, signal in this sparse network configuration.

Hybrid Model:

- Superiority of Hybrid Approach: The additive hybrid model (HYB_AA) achieves AUC equals 0.5316 ± 0.0046, outperforming both the best local heuristic (AA equals 0.5221 ± 0.0045) and standalone community detection (Louvain equals 0.5270 ± 0.0045). The improvement of delta equals 0.0095 AUC over pure AA represents a 2.1 sigma gain, confirming statistical significance given the small standard deviations.
- This result validates the complementary nature of multi-scale signals. The performance gain stems directly from the zero-score rescue mechanism: 97.7 percent of test edges have no common neighbors, where local methods assign score equals 0 and cannot discriminate. The hybrid model leverages community membership to assign non-zero scores (alpha equals 2.0) to same-community pairs in this regime, dramatically improving recall. Community detection identifies the broad research field, while local heuristics pinpoint specific collaborative partners within those fields. The additive formulation successfully synthesizes these two layers of information.

4.3 Qualitative Results

To validate the practical utility of the system beyond aggregate metrics, we conducted a case study using the interactive prototype.



meth…	rank	candidate	score	common_neighbor_names	shortest_path
AA	1	A. Shapere	1.1162212531024944	M. R. Plesser, P. C. Argyres	E. Witten → M. R. Plesser → A. Shapere
AA	2	A. Hanany	0.9040868828124409	M. R. Plesser, N. Seiberg	E. Witten → M. R. Plesser → A. Hanany
CN	1	A. Hanany	2	M. R. Plesser, N. Seiberg	E. Witten → M. R. Plesser → A. Hanany
CN	2	A. Shapere	2	M. R. Plesser, P. C. Argyres	E. Witten → M. R. Plesser → A. Shapere
HYB_AA	1	A. Shapere	1.1162212531024944	M. R. Plesser, P. C. Argyres	E. Witten → M. R. Plesser → A. Shapere
HYB_AA	2	A. Hanany	0.9040868828124409	M. R. Plesser, N. Seiberg	E. Witten → M. R. Plesser → A. Hanany
PA	1	S. Ferrara	342	C. Vafa	E. Witten → C. Vafa → S. Ferrara

candidate	n_methods	methods
A. Shapere	4	AA, CN, HYB_AA, RA
A. Hanany	3	AA, CN, HYB_AA
E. Zaslow	1	RA
S. Ferrara	1	PA
W. Lerche	1	PA

Figure 2: Illustrates the recommendation results for the query author "E. Witten", a renowned physicist in the dataset.

Analysis of Observations:

- Successful Structural Detection: The system identifies A. Shapere and A. Hanany as the top candidates across multiple methods (AA, CN, Hybrid).
- Explainability: Unlike black-box predictions, the interface provides transparent justifications. For the top candidate A. Shapere:
  - Common Neighbors: It explicit lists the “bridge” authors: M. R. Plesser, P. C. Argyres.
  - Shortest Path: It visualizes the connection path: E. Witten → M. R. Plesser → A. Shapere, allowing the user to trace the academic lineage.
- Consensus vs. Diversity: The “Agreement” panel on the right highlights that A. Shapere is recommended by 4 different methods (AA, CN, HYB_AA, RA), signaling a high-confidence prediction. Interestingly, Preferential Attachment (PA) suggests a completely different candidate (S. Ferrara), leveraging "hub" status rather than shared neighbors. This contrast vividly demonstrates the system's ability to offer diverse perspectives on potential collaboration.

Conclusion: This qualitative inference proves that the system functions effectively as a discovery tool, allowing researchers to explore potential partnerships based on verifiable network connections.

4.4 Empirical Analysis

4.4.1 Comparative Analysis & Ablation

To isolate the contribution of different structural signals, we systematically evaluate individual heuristics versus composite models.

- Local Proximity: Classical neighbor-based methods (CN, AA, RA) consistently achieve stable performance ($AUC \approx 0.52$), outperforming random guessing. This confirms that local triadic closure is a valid predictor.

- Global Popularity: A critical finding is the underperformance of Preferential Attachment (PA) ($AUC \approx 0.19$), which performs significantly worse than random. This "anti-preferential" outcome indicates that in this specific temporal period of theoretical physics, new collaborations were not driven by attachment to high-degree hubs, but potentially by niche, specialized interactions.

- Hybrid Synergy: The hybrid models consistently outperform their single-method counterparts ($AUC \approx$ 0.53 vs 0.52). This indicates a positive synergy: while Local methods are precise but limited to close neighbors, the Global Community signal helps bridge the gap for authors who are topologically distant but thematically related.

- This result validates the complementary nature of multi-scale signals. The performance gain stems directly from the zero-score rescue mechanism: 97.7 percent of test edges have no common neighbors, where local methods assign score equals 0 and cannot discriminate. The hybrid model leverages community membership to assign non-zero scores (alpha equals 2.0) to same-community pairs in this regime, dramatically improving recall. Community detection identifies the broad research field, while local heuristics pinpoint specific collaborative partners within those fields. The additive formulation successfully synthesizes these two layers of information.

4.4.2 Failure Analysis

While effective for discovery, the system exhibits distinct failure modes:

- Community False Positives: The community-based module (and by extension, the hybrid model) risks assigning high scores to author pairs solely because they reside in the same large partition (e.g., “String Theory”), even if they have no direct research overlap.

- Sparsesness Sensitivity: Proximity methods (CN/AA) inherently fail to predict “long-range” collaborations where two authors have zero mutual friends. In our sparse graph, these “cold-start” links are common, leading to false negatives (scores of 0).

- Hub Bias: Although PA generally performed poorly, it still tends to indiscriminately recommend “stars” (e.g., E. Witten) to everyone. In cases where a young researcher does collaborate with a star, PA might correctly predict it, but at the cost of thousands of false recommendations to others.

4.4.3 Error Analysis

An inspection of misclassified links reveals that:

- False Negatives (Missed Links) often involve authors with low degrees ($k < 3$). In these cases, the topological "footprint" is too faint for heuristic methods to detect a signal.
- False Positives (Hallucinations) frequently occur in dense cliques where triadic closure is high, but social factors (e.g., competing research groups) prevent actual collaboration - a nuance that purely structural graph metrics cannot perceive.

4.4.4 Robustness Check

All algorithms are deterministic given the graph structure. To ensure the reliability of the observed "anti-PA" phenomenon, we verified the consistency of the results across the full test set (over 10,000 positive edges) rather than a small subset. The rigorous time-aware split ensures these findings reflect genuine temporal dynamics rather than sampling artifacts.

5. Discussion

Interpretation of Results: Contrary to traditional network theory expectations, our analysis reveals a distinct "anti-preferential attachment" regime ($AUC \approx 0.19$). The data suggests that new collaborations were not driven by the popularity of "hub" researchers; instead, scientists favored connecting with peers of specific relevance rather than accumulating ties with the most connected individuals. Critically, our optimized additive Hybrid model (AUC ≈ 0.53) successfully outperformed both local heuristics (AUC ≈ 0.52) and standalone community detection. This demonstrates that global community structure and local neighborhood topology are complementary signals. While local methods offer high precision for close connections, the community signal provides essential 'macro-level' context, enabling the discovery of potential collaborations between topologically distant but thematically related researchers.

Achievement of Objectives: Despite the challenging nature of the prediction task, the project successfully fulfilled its core engineering and design goals. We established a robust time-aware evaluation pipeline, preventing common methodological pitfalls like data leakage. Most notably, the development of the interactive prototype bridges the gap between abstract metrics and practical utility. By rendering transparent justifications (shared neighbors and academic lineage paths), the system transforms a standard classification problem into an explainable discovery tool, directly addressing the project's emphasis on interpretability.

Critical Challenges Encountered:

- Data Heterogeneity: The raw ArXiv data contained significant inconsistencies (e.g., LaTeX formatting, embedded affiliations). Developing a robust regex-based normalization pipeline was essential to prevent node duplication (e.g., treating "H. Dahmen" and "H. D. Dahmen" as distinct).

- Network Sparsity & Cold-Start: With an average degree of only 3.67, the network is highly sparse. This posed a major challenge for local heuristics, as many author pairs share no common neighbors, resulting in zero-gradients and limiting predictive recall.

- Scalability: Computing path-based metrics (like Shortest Path for explanation) for all pairs is computationally prohibitive ($O(N^3)$). We overcame this by restricting real-time explanations to only the Top-K candidates in the prototype.

6. Conclusion and Future Work

6.1 Conclusion

This study investigated the problem of co-author recommendation in scientific networks by systematically benchmarking classical structural heuristics against a rigorous temporal evaluation framework. Using the ArXiv Hep-Th dataset, we constructed a weighted collaboration graph and developed a suite of predictive models ranging from local proximity (CN, AA) to community-aware hybrid approaches.

Our experimental findings highlight the complexity of real-world collaboration dynamics:

- Heuristic Efficacy & Hybrid Enhancement: While local proximity methods (CN, AA) provided stable baselines (AUC ≈ 0.52), their discriminative power was constrained by network sparsity. However, our Hybrid Framework successfully mitigated this limitation by integrating community structure, achieving a superior AUC of 0.53. This results confirms that “triadic closure” works best when augmented with broader “community relevance”.

- Anti-Preferential Trend: Notably, the study revealed a counter-intuitive distinct failure of Preferential Attachment ($AUC < 0.2$), indicating that in this specific domain and era, collaboration was not governed by a "rich-get-richer" mechanism.

- Interpretability Contribution: Beyond quantitative metrics, the project successfully delivered a functional interactive prototype powered by Gradio. By effectively visualizing justification paths and shared neighbors, the system fulfilled its primary objective of providing an explainable, user-centric discovery tool.

6.2 Future Work

Semantic Integration: The current structural approach misses the “context” of research. Parsing paper abstracts and keywords to compute semantic similarity (e.g., via TF-IDF or BERT embeddings) would likely significantly reduce false positives by filtering structurally close but topically distant pairs.

Advanced Representation Learning: Moving beyond manual heuristics, applying Graph Neural Networks (GNNs) (e.g., GraphSAGE, GAT) could automatically learn higher-order node embeddings that capture complex structural motifs missed by 2-hop metrics.

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[7] Link Property Prediction (OGB collaboration dataset)

Appendices

A. Technical Details

Code Repo

- Link Github
- Link Raw Dataset
- Link Clean Data
- Link Data Mining
- Link Main Code
- Link Slide
- Link Final Report

Hyperparameter Settings

Parameter	Value	Description
Hybrid Alpha (α)	2.0	Additive bias weight (Selected via Grid Search).
Community Resolution	1.0	Standard resolution for Louvain Algorithm.
Explanation Depth	<= 4 hops	Cutoff limit for Shortest Path search.
Top-K	10	Number of recommendations presented to users.
Split Year (T)	1998	Temporal boundary (Train <= 1998 < Test).

Data Sample

- Sample rows from authors.csv (Node Metrics)

author,first_year,n_papers,affiliations_seen

A C Tort,2001,1,[]

A Calogeracos,2000,1,[]

A M Semikhatov,1992,15,"['Lebedev Physics Institute', 'PN Lebedev Institute']"

A. A. Andrianov,1993,13,"['Institute of Physics, Sankt-Petersburg']"

A. A. Tseytlin,1992,121,[]

- Sample rows from coauthor_edges.csv (Edge List)

author1,author2,first_year,weight

A C Tort,F C Santos,2001,1

A. A. Andrianov,M. V. Ioffe,1993,5

A. A. Belavin,R. A. Usmanov,2000,2

A. A. Bytsenko,S. D. Odintsov,1992,12

A. A. Tseytlin,I. R. Klebanov,1996,10

B. Project Planning

Timeline

- Weeks 1–2: Literature review, dataset exploration

- Weeks 3–4: Metadata parsing, graph construction
- Weeks 5–6: Implementation of heuristics, community detection
- Week 7: Temporal split & evaluation
- Week 8: Hybrid model + UI prototype
- Week 9: Experiments, results & analysis
- Week 10: Report writing & presentation preparation

Team Responsibilities

Ha Nhat Thai:

- Data extraction & cleaning
- Graph construction & heuristics implementation
- Write Report

Nguyen Khanh Tuan:

- Evaluation & hybrid model
- UI prototype
- Create slide